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| **Fiscal Year:** | FY 2024 | **Task Last Updated:** | FY 09/19/2023 |
| **PI Name:** | Shippen, Dorothy Ph.D. | | |
| **Project Title:** | Telomere Dynamics and Oxidative Stress in Arabidopsis in the Space Radiation Environment | | |
| **Division Name:** | Space Biology | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | SPACE BIOLOGY--Cellular and molecular biology<br>SPACE BIOLOGY--Radiation biology | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 77843 | **Congressional District:** | 10 |
| **Comments:** | | | |
| **Project Type:** | Flight,Ground | **Solicitation / Funding Source:** | 2021 Space Biology NNH21ZDA001N-SBPS E.9: Plant Studies |
| **Start Date:** | 11/21/2022 | **End Date:** | 11/20/2025 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | 3 | **Monitoring Center:** | NASA KSC |
| **Contact Monitor:** | O'Rourke, Aubrie | **Contact Phone:** | |
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| **Flight Program:** | | | |
| **Flight Assignment:** | ISS | | |
| **Key Personnel Changes/Previous PI:** | No changes | | |
| **COI Name (Institution):** | Bailey, Susan Ph.D. ( Colorado State University )<br>Wyatt, Sarah Ph.D. ( Ohio University ) | | |
| **Grant/Contract No.:** | 80NSSC23K0302 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | We propose an International Space Station (ISS) Flight Research Investigation using Spectrum hardware, and radiation experiments at NASA Space Radiation Lab (NSRL, Brookhaven) and Colorado State University. This proposal leverages the expertise of three international leaders in the fields of telomere, radiation, and plant biology to explore the relationship between the space radiation environment, oxidative stress, and the dynamics of plant telomeres. Our preliminary data show that, in response to spaceflight, telomerase enzyme activity in Arabidopsis is strongly induced, but telomere length is steady. Telomerase activity was not substantially elevated by simulated microgravity conditions, implying that space radiation is responsible for telomerase induction. Our central hypothesis is that exposure to the space radiation environment results in elevated Reactive Oxygen Species (ROS) which in turn leads to genome oxidation and the high-telomerase activity phenotype observed in the telomere maintenance machinery of space flown plants. We will test this hypothesis with three specific Objectives. Objective 1 will examine the effect of the combined stressors associated with spaceflight on plant telomere dynamics, genome oxidation, and organellar stability in Arabidopsis seedlings grown aboard the ISS as compared to ground controls. We will employ a series of mutant and overexpression lines that influence telomeres and genome oxidation. Objective 2 will explore the impact of space radiation exposure on these parameters using the same Arabidopsis lines by studying the effect of fractionated schemes of simulated space radiation exposure (Galactic Cosmic Ray Simulator at the NSRL) as well as low dose, low dose-rate neutron and low dose gamma-ray exposure (irradiation facilities at Colorado State University). In Objective 3, we synthesize the data gathered from spaceflight and on-ground fractioned radiation exposure experiments to address our central hypothesis. This work is highly impactful as telomeres are an important biological marker of survivability in both plants and humans. Elucidating the relationship between the space radiation environment and telomere dynamics may prove to be a critical component of long-duration missions in the future. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | This research addresses how plants endure the impact of the space radiation environment. Our experiments may provide new insight into how different stressors alter the plant genome and the oxidative stress response, and the type of genetic modifications that can be made to enhance survival in harsh environments, including those that currently exist on Earth. |
| **Task Progress:** | SCIENTIFIC GOALS AND OBJECTIVES<br>Telomeres are an important biological marker of survivability in all eukaryotes. In this project, Principal Investigator (PI) D. Shippen and Co-Principal Investigators (Co-PIs) S. Wyatt and S. Bailey combine expertise and resources to assess the impact of the space radiation environment on oxidation status, telomere length dynamics, and genome stability in Arabidopsis. Preliminary data gathered from a previous spaceflight and ground experiments showed a dramatic induction of telomerase enzyme activity; yet telomere length remained unchanged in space-flown plants. Notably, telomerase activity was not substantially elevated under simulated microgravity conditions on the ground, implying that radiation or some combined stressors in spaceflight are responsible for telomerase induction. The central hypothesis of this study is that exposure to the space radiation environment results in elevated levels of reactive oxygen species (ROS), which in turn leads to genome oxidation and the high-telomerase activity phenotype observed in the telomere maintenance machinery of space-flown plants.<br><br>To test this hypothesis, three Objectives were proposed. For Objective 1, PI Shippen and Co-PI Wyatt will test the effect of spaceflight on telomere length, telomerase activity, and oxidative stress in International Space Station/ISS-grown Arabidopsis mutants with altered telomere maintenance machinery or altered redox metabolism. The plants will bear a transgenic ROS-reporter construct and be grown in Spectrum hardware. The fluorescence microscopy capability of Spectrum will be employed to genetically monitor ROS induction during spaceflight. Space-flown plants will be subjected to biochemical and molecular-based assays to measure the impact of space radiation on redox homeostasis, genome stability, organelle integrity, telomere length, and telomerase enzyme activity. In Objective 2, the research teams of PI Shippen and Co-PI Wyatt will team up with Co-PI Bailey to conduct ground studies to assess the influence of space radiation exposure on these same parameters using the Galactic Cosmic Ray Simulator at the NASA Space Radiation Lab at Brookhaven National Lab, and the low dose, low dose-rate irradiation facilities at Colorado State University. The goal is to establish dose responses for plant telomere dynamics, genome stability and oxidation, and organellar function in ground/Earth-grown Arabidopsis. Objective 3 is synthesis of data gathered from spaceflight and on-ground fractioned radiation exposure experiments to evaluate the central hypothesis.<br><br>SUMMARY OF THE PAST YEAR'S RESEARCH ACCOMPLISHMENTS<br><br>Pre-Flight, Ground Experiments: Our primary focus during this first year has been to construct the required genetic lines necessary for flight and ground radiation experiments. We have made good progress on the following goals:<br><br>PF.1.) Generate ROS-sensing reporters in different genetic backgrounds (Shippen and Wyatt labs).<br><br>Zat12p is a transcription factor that is activated by oxygen radicals. We obtained the Zat12p-ROS transgenic line from Dr. Won-Gyu Choi. Seeds from this line were used to generate ROS-sensing reporters in different genetic backgrounds. Crosses were made independently in the Shippen (Postdoctoral researcher Borja Barbero Barcenilla) and Wyatt (Graduate student Emma Canaday) labs. Specifically, we crossed plants bearing Zat12p-ROS with lines bearing a mutation in genes implicated in the oxidative stress response (cat2 and ogg1) and in telomere length maintenance (tert and ten1). Following genetic crosses, polymerase chain reaction (PCR) genotyping was performed to identify homozygous mutants carrying the Zat12p-ROS reporter. These plants were grown to bulk up seed stock for all pre-flight and flight experiments. To confirm that CAT2 was indeed over-expressed in the CAT2 OE Zat12-ROS line, western blotting was performed using an AtCAT2 specific antibody. Several transgenic lines were identified that express high levels of CAT2.<br><br>PF.2.) Assess ROS induction (Shippen and Wyatt labs).<br><br>To validate the activity of our ROS reporter lines, the Shippen and Wyatt labs independently examined seedlings of a variety of different genetic backgrounds using fluorescent microscopy. As expected, fluorescence signal was detected in the Zat12-ROS reporter lines, but not in the wildtype Col-0 control. Fluorescence was detected in the reporter lines in seedlings grown in agar plates and directly imaged (Wyatt lab) or following transfer of the seedling to a glass slide (Shippen lab). We plan to quantitate ROS induction in these lines by treating them with H2O2 and then measuring ROS intensity by fluorescence microscopy. To further validate our results, we will compare our microscopy-based fluorescence measurements with a standard spectrophotometric technique specifically designed to quantify H2O2 |

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| | (Junglee et al., 2014 Am. J. Anal. Chem. 5:730). [Ed. Note: See References.]<br><br>REFERENCES<br><br>Junglee S, Urban L, Sallanon H, Lopez-Lauri F. Optimized assay for hydrogen peroxide determination in plant tissue using potassium iodide. American Journal of Analytical Chemistry. 2014;5(11):730. <a target="_blank" href="http://dx.doi.org/10.4236/ajac.2014.511081">http://dx.doi.org/</a> |
| **Bibliography Type:** | Description: (Last Updated: 06/03/2026) |
| **Abstracts for Journals and Proceedings** | Barbero Barcenilla, B. "Spaceflight and microgravity response of plant telomeres and telomerase." Annual Meeting of the Southern Section of American Society of Plant Biologists, Fayetteville, AR, March 25-27, 2023.<br>Abstracts. Annual Meeting of the Southern Section of American Society of Plant Biologists, Fayetteville, AR, March 25-27, 2023. , Mar-2023 |
| **Abstracts for Journals and Proceedings** | Barbero Barcenilla, B. "Effect of spaceflight, lunar and Martian regolith on plant telomeres and telomerase." International Space Development Conference (ISDC), Frisco, TX, May 25-28, 2023.<br>Abstracts. International Space Development Conference (ISDC), Frisco, TX, May 25-28, 2023. , May-2023 |
| **Abstracts for Journals and Proceedings** | Barcenilla BB, Meyer AD, Castillo-González C,Young P, Min J-H, Song J, Phadke C, Land E, Canaday E, Perera IY, Bailey SM, Aquilano RO, Wyatt S, Shippen DE. "Telomerase takes off: Uncoupling telomerase enzyme activity from telomere length maintenance in space." 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023.<br>Abstracts. 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. , Nov-2023 |
| **Abstracts for Journals and Proceedings** | Barbero Barcenilla B. "Spaceflight and microgravity response of plant telomeres and telomerase." 8th Edition of Global Conference on Plant Science and Molecular Biology" (GPMB 2023), Valencia, Spain/Virtual, September 11-13, 2023.<br>Abstracts. 8th Edition of Global Conference on Plant Science and Molecular Biology" (GPMB 2023), Valencia, Spain/Virtual, September 11-13, 2023. , Sep-2023 |
| **Abstracts for Journals and Proceedings** | Barbero Barcenilla B. "Spaceflight and microgravity response of plant telomeres and telomerase." 38th Annual Meeting of the American Society for Gravitational and Space Research, Houston, TX, November 9-12, 2022.Poster Session.<br>Abstracts. 38th Annual Meeting of the American Society for Gravitational and Space Research, Houston, TX, November 9-12, 2022. , Nov-2022 |
| **Abstracts for Journals and Proceedings** | Barbero Barcenilla B, Meyers A, Castillo-Gonzalez C, Land E, Perera I, Aquilano R, Canaday E, Wyatt S, Shippen D. "Spaceflight and microgravity response of plant telomeres and telomerase." Forming and Exploring Habitable Worlds Conference (University of Edinburgh), Edinburgh Scotland, November 7-13, 2022.<br>Abstracts. Forming and Exploring Habitable Worlds Conference (University of Edinburgh), Edinburgh Scotland, November 7-13, 2022. , Nov-2022 |
| **Articles in Other Journals or Periodicals** | Barcenilla BB, Meyer AD, Castillo-González C,Young P, Min J-H, Song J, Phadke C, Land E, Canaday E, Perera IY, Bailey SM, Aquilano RO, Wyatt S, Shippen DE. "Arabidopsis telomerase takes off by uncoupling enzyme activity from telomere length maintenance in space." Nature Communications, in press as of October 2023. , Nov-2023 |